SBI 812

Machine Learning in Bioinformatics

Course Outline | Master of Science in Bioinformatics

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Course duration	10 teaching weeks	**Contact time**	3 hours per week (30 hours)
Entry level	No reliable programming or ML prerequisites	**Core technologies**	Python, pandas, visualization and scikit-learn
Data focus	Mixed biological datasets	**Course approach**	Hands-on, reproducible and AI-integrated

1. Course description

This course introduces machine-learning concepts and their application to biological data. It begins with data handling and visualization in Python, then develops practical understanding of supervised and unsupervised learning, model evaluation, data leakage, interpretation and reproducibility. Examples are drawn from mixed biological datasets, including gene expression, sequence-derived features, microbiome profiles and phenotypic measurements.

The emphasis is not on memorizing algorithms. Students learn to define an appropriate prediction or discovery problem, prepare data correctly, establish a defensible baseline, select suitable evaluation procedures and distinguish statistical performance from biological usefulness.

AI may help build a model, but the student must justify the data, evaluation and biological interpretation.

2. Course purpose

The course is designed for MSc students whose prior programming and statistics preparation may vary. Foundational skills are introduced within the course, and each new method is taught through an interpretable biological example before more complex workflows are attempted.

3. Learning outcomes

1. Load, inspect, clean and summarize biological data using pandas.
2. Visualize distributions, relationships, missingness and class imbalance.
3. Distinguish features, outcomes, training data, validation data and test data.
4. Explain regression, classification, clustering and dimensionality reduction at an introductory level.
5. Construct reproducible scikit-learn workflows for biological datasets.
6. Select evaluation metrics appropriate to the biological question and data structure.
7. Recognize overfitting, data leakage, confounding and misleading performance estimates.
8. Compare baseline, linear, tree-based and ensemble models critically.
9. Interpret model outputs without overstating biological or causal conclusions.

10. Use AI to support analysis while verifying code, assumptions, results and claims.

4. Teaching and learning approach

Each session combines accessible concepts, a live worked example, guided analysis and an AI-assisted diagnostic challenge. Students work with small, interpretable datasets before progressing to a capstone analysis. Mathematical notation is introduced only where it clarifies the behaviour of a method.

Component	Approximate time
Concepts and biological framing	45 minutes
Instructor demonstration	30 minutes
Guided practical	70 minutes
AI-assisted model audit	25 minutes
Review and formative assessment	10 minutes

5. Weekly course schedule

Week	Topic	Core content	Practical outcome
1	Python and pandas foundations	DataFrames, columns, data types, indexing, filtering, summaries and missing values.	Load and inspect a biological dataset reliably.
2	Exploration and preprocessing	Visualization, distributions, scaling, encoding, outliers, missingness and biological metadata.	Create a defensible exploratory and preprocessing report.
3	Machine-learning workflow	Features, outcomes, train/test split, baselines, pipelines, overfitting, leakage and reproducibility.	Build a leakage-aware baseline workflow.
4	Regression	Continuous outcomes, linear regression, residuals, MAE, RMSE and R-squared.	Predict a continuous biological phenotype and evaluate errors.
5	Classification	Logistic regression, confusion matrix, sensitivity, specificity, precision, recall, ROC-AUC and PR-AUC.	Classify biological samples using suitable metrics.
6	Trees and ensembles	Decision trees, random forests, feature importance, tuning and overfitting control.	Compare interpretable and ensemble classifiers.
7	Clustering	Distances, k-means, hierarchical clustering, cluster stability and biological interpretation.	Explore groups in an unlabeled biological dataset.
8	Dimensionality reduction	PCA, variance, loadings, visualization and the limits of two-dimensional projections.	Visualize high-dimensional omics-style data responsibly.
9	Evaluation and interpretation	Cross-validation, imbalance, tuning, model comparison, permutation importance and interpretation limits.	Audit and improve an AI-assisted model workflow.
10	Integrated capstone and defence	Problem formulation, reproducible modelling, results, limitations, biological claims and AI disclosure.	Complete and defend a mixed-dataset ML project.

6. Assessment strategy

Assessment rewards correct problem formulation, careful data handling and defensible evaluation rather than model complexity alone.

Assessment	Weight	Purpose
Weekly practical exercises	20%	Progressive practice in data handling, modelling and evaluation.
Independent data-analysis practical	20%	Essential individual competence in pandas, interpretation and basic modelling.
AI-generated model audit	15%	Detection and correction of leakage, metric misuse, weak validation and unsupported claims.
Machine-learning capstone project	35%	Reproducible analysis of a mixed biological dataset with justified methods and limitations.
Oral project defence	10%	Explanation and modification of code, evaluation choices and biological conclusions.

Assessment modes

Mode	AI rule	Educational purpose
A - Independent	No generative AI	Demonstrate essential data and model literacy.
B - AI-supported	AI for explanation, code suggestions and debugging	Develop critical supported learning.
C - Open AI	Full use with disclosure, testing and verification	Simulate responsible professional practice.

7. Capstone project

Each student or approved small group will analyse a supplied biological dataset containing a clearly defined outcome or exploratory question. Dataset choices may include gene-expression measurements, sequence-derived features, microbiome abundance profiles or combined biological and phenotypic variables.

A valid submission must include the question, data dictionary, exploratory analysis, preprocessing decisions, baseline, model comparison, evaluation procedure, interpretation, limitations, reproducible code and AI-use declaration. A simpler well-evaluated model is preferred to an unnecessarily complex model.

8. Responsible AI-use policy

Generative AI may be used only at the level specified for each activity. Students remain responsible for data handling, code, model selection, evaluation, results and interpretation. Material AI assistance must be disclosed, and students must be able to explain and modify submitted work during an oral or practical assessment.

Students must verify that AI-generated workflows do not leak outcome information, split related samples incorrectly, select metrics after seeing test results or make causal and biological claims unsupported by the analysis. Confidential, identifiable, clinical or unpublished data must not be uploaded to unauthorized AI services.

9. Learning resources and software

- Windows Subsystem for Linux 2 with Ubuntu 24.04 LTS
- Anaconda Distribution and Python
- pandas, NumPy, Matplotlib, Seaborn and scikit-learn
- Git and GitHub course repository
- Instructor-provided mixed biological datasets and data dictionaries